BigBird: Transformers for Longer Sequences

A short research overview

Topic: Natural Language Processing (NLP) and efficient attention in Transformer models

Research question
How does BigBird help Transformer models work with longer text?

This report gives a brief introduction to BigBird, explains its main idea, compares its attention approach with BERT's standard full attention, and summarizes common use cases and limitations.

Based mainly on the original NeurIPS 2020 paper by Zaheer et al.

1. Introduction

BigBird is a Transformer architecture introduced in a research paper published at NeurIPS 2020. It was designed to make it easier to process long sequences of text. Models such as BERT use self-attention, which lets each token consider other tokens in the input. With full attention, the amount of attention computation and memory can grow quadratically as the sequence becomes longer. This can make long documents expensive to process.

BigBird addresses this issue by using **sparse attention**. Instead of connecting every token to every other token, it uses a carefully selected set of attention connections. This reduces the cost of attention while still allowing information to travel across the sequence.

2. How BigBird's Attention Works

The paper combines three main kinds of attention connections:

- **Local attention:** a token attends to nearby tokens. This helps the model capture information from surrounding words.
- **Random attention:** some connections are made to randomly selected tokens. These connections can help information move between distant parts of the text.
- **Global attention:** selected global tokens can attend to the whole sequence and can be attended to by other tokens. This gives the model a way to collect and share information across the document.

The exact attention pattern depends on the implementation and configuration. The key idea is that the model uses fewer connections than full attention, but chooses them to preserve useful information flow.

3. BigBird and BERT

Point	Standard full attention (e.g., BERT)	BigBird sparse attention
Attention connections	Each token can attend to every token in the input.	Each token attends to selected local, random, and/or global connections.
Scaling with sequence length	Attention cost grows quadratically with sequence length.	Can scale linearly with sequence length when block size and the number of global tokens are fixed.
Main trade-off	Simple attention pattern, but costly for long inputs.	More efficient for long inputs, but uses a more structured attention pattern.

The linear-scaling statement refers to the sparse attention operation under the paper's assumptions; it does not mean every part of every BigBird system is always linear in every setting.

4. Where Can BigBird Be Useful?

BigBird can be useful for NLP tasks where relevant information is spread across a long input. Examples discussed in the paper and related model work include:

- **Question answering:** answering questions using long passages or documents.
- **Document classification:** predicting a category for a report or other long text.
- **Summarization:** using a longer source document when generating a shorter summary.

- **Other long-sequence tasks:** the paper also explores applications such as genomics, where the input is a long sequence.

5. Advantages and Limitations

Advantages

- It can handle longer input sequences than standard full-attention configurations under comparable hardware constraints.
- Its sparse attention reduces the attention computation and memory requirements as sequence length grows.
- The local, random, and global connections provide different ways for information to move through the sequence.

Limitations

- The sparse attention pattern is more complex to implement and configure than full attention.
- Performance depends on the task, data, and model configuration; sparse attention is not automatically better for every problem.
- Reducing attention cost does not remove all computational costs associated with training or running a large Transformer.

6. Conclusion

BigBird is an approach for making Transformer models more practical for long text. Its main idea is to replace full attention with a sparse pattern that combines local, random, and global connections. This can reduce the attention cost from quadratic to linear in sequence length under fixed-pattern assumptions, making BigBird useful for tasks such as question answering and summarization. The main trade-off is that the attention pattern is more complicated, and the best choice still depends on the application.

7. References

Zaheer, M., Guruganesh, G., Dubey, A., Ainslie, J., Alberti, C., Ontañón, S., Pham, P., Ravula, A., Wang, Q., Yang, L., & Ahmed, A. (2020). *Big Bird: Transformers for Longer Sequences.* Advances in Neural Information Processing Systems (NeurIPS 2020).

Official paper page: https://proceedings.neurips.cc/paper/2020/hash/c8512d142a2d849725f31a9a7a361ab9-Abstract.html

Google Research publication page: https://research.google/pubs/big-bird-transformers-for-longer-sequences/

Official implementation: https://github.com/google-research/bigbird